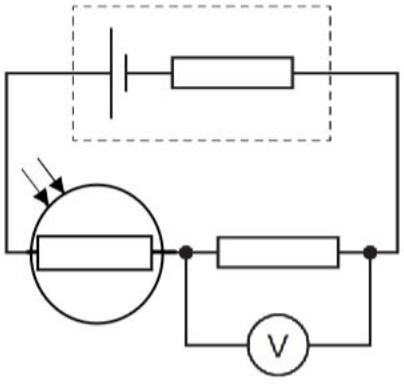
21 A cell with internal resistance is connected to a light-dependent resistor (LDR), a fixed resistor and a voltmeter as shown.



The voltmeter reading increases.

Which quantity **decreases** as the voltmeter reading increases?

A the charge moving through the cell per unit time

B the energy transferred to the fixed resistor per unit charge

C the intensity of the light incident on the LDR

D the terminal potential difference across the cell

Ans: D

When pd across voltmeter increases, this suggest that the current through the fixed resistor increases, since R of fixed resistor is constant. ($V = IR$)

By $E = V + Ir \rightarrow V = E - Ir$, a larger I, for the same emf E, the terminal pd V **must decrease**.

Cannot be A, since current increase, the charge moving through cell must also increase.
Cannot be B, since energy per unit charge to the fixed resistor is p.d. and p.d. increased.
Cannot be C, since for pd across fixed R to increase, the main current must increase. Thus the resistance of LDR must decrease. When light intensity decreases, resistance of LDR would increase. A contradiction.